

In [17]:

In [27]:

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import tkinter as tk
from tkinter import messagebox
import random
import string

# Function to generate password
def generate_password():
    try:
        length = int(entry_length.get())
        if length < 4:
            raise ValueError("Password length must be at least 4.")

        # Ensure at least one character from each category
        password = [
            random.choice(string.ascii_uppercase),
            random.choice(string.ascii_lowercase),
            random.choice(string.digits),
            random.choice(string.punctuation)
        ]

        # Fill the rest of the password length with random characters
        if length > 4:
            characters = string.ascii_letters + string.digits + string.punctuation
            password += [random.choice(characters) for _ in range(length - 4)]

        # Shuffle the list to ensure randomness and convert to a string
        random.shuffle(password)
        password = ''.join(password)

        # Display the generated password
        label_password.config(text=f"Generated Password: {password}", fg="green")
    except ValueError as e:
        messagebox.showerror("Invalid Input", str(e))

# Set up the main window
root = tk.Tk()
root.title("Password Generator")

# Create and place the widgets
label_length = tk.Label(root, text="Password length:")
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label_length.grid(row=0, column=0, padx=10, pady=10)

entry_length = tk.Entry(root)
entry_length.grid(row=0, column=1, padx=10, pady=10)

button_generate = tk.Button(root, text="Generate Password", command=generate_password, bg="green", fg="white")
button_generate.grid(row=1, column=0, columnspan=2, pady=10)

label_password = tk.Label(root, text="Generated Password:")
label_password.grid(row=2, column=0, columnspan=2, padx=10, pady=10)

# Run the application
root.mainloop()
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In []: